MATH 101 Unit 1: Expressions, Equations, and Inequalities - Practice 1

Version 202610122020.34 | Name: _____

Complete and show all work on a separate sheet of paper. Record final answers on the answer sheet. Give exact answers and simplify completely. Only a basic four-function calculator may be used. Suggested timed practice: 30 minutes.

Answer key follows the answer sheet. Attempt the test before checking it.

1. Solve the inequality and write the solution in interval notation.

$|2x - 1| \le 3$

2. Simplify and state all values excluded by the original expression.

$$\frac{x^2 - 4}{x^2 + 2x - 8}$$

3. Simplify. In (a), x and y are nonzero; give positive exponents.

(a) $\frac{6x^{-2}y^3}{3xy^{-1}}$

(b) $\sqrt{18}$

4. Solve both equations. In (b), state original restrictions and check your answer.

(a) $x^2 + 2x - 3 = 0$

(b) $\frac{4}{2x-4} + \frac{1}{x-2} = 3$

5. (a) Expand and simplify. (b) Factor completely.

(a) $(2x - 3)(x + 2)$

(b) $2x^2 - 5x - 3$

Student answer sheet

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Question 1

Derived bounds

12 pts

Interval

8 pts

Question 2

Expression

14 pts

Excluded values

6 pts

Student answer sheet

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Question 3

(a)

10 pts

(b)

10 pts

Question 4

(a)

10 pts

(b), with check

10 pts

Question 5

(a)

10 pts

(b)

10 pts

Answer key

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Question 1

Derived bounds	$-1 \le x \le 2$	12 pts
Interval	$[-1, 2]$	8 pts

Question 2

Expression	$\frac{x+2}{x+4}$	14 pts
Excluded values	$\{-4, 2\}$	6 pts

Answer key

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Question 3

(a) $\frac{2y^4}{x^3}$ 10 pts

(b) $3\sqrt{2}$ 10 pts

Question 4

(a) $x = -3,\ 1$ 10 pts

(b), with check $x = 3;\ x \neq 2$ 10 pts

Question 5

(a) $2x^2 + x - 6$ 10 pts

(b) $(x - 3)(2x + 1)$ 10 pts